

Implementing Zero Trust with the Power of Open Source

Zero trust architecture that trusts nothing and no one is the order of the day.

What's great is that open source offers a range of tools to implement such architecture. Find out what these tools are, and how to put them to best use.



The traditional security model – a hardened perimeter protecting a trusted internal network – is crumbling. Cloud adoption, remote workforces, and sophisticated threats render the ‘castle-and-moat’ approach obsolete. As a result, zero trust architecture (ZTA), a paradigm shift demanding ‘never trust, always verify’ is becoming popular day by day. It assumes that threats can come from both outside and inside the network. Every request must be authenticated, authorised, and encrypted before being granted access—regardless of its origin.

The principle is simple: treat every user, device, and network flow as potentially hostile, requiring continuous authentication and authorisation. While often associated with expensive enterprise suites, ZTA is powerfully achievable using robust open source solutions.

Various open source software used to implement ZTA

Implementing ZTA rests on key pillars, each addressable with the kind of mature open source tools listed here.

- **Keycloak** from Red Hat is the powerhouse. It provides single sign-on (SSO), multi-factor authentication (MFA), identity federation (SAML, OIDC), and fine-grained authorisation (via policies).

- **Pomerium** is an identity-aware reverse proxy that provides secure access to internal applications without the need for corporate VPN. Pomerium integrates with various identity providers and enforces policies, ensuring that only authorised users can access specific resources.

- **Osquery** by Facebook/Meta is revolutionary. It exposes operating system data as a SQL database, enabling powerful queries to assess device state (patches, encryption, running processes).
- **Cloudflare Access** is part of Cloudflare’s zero trust platform and replaces traditional VPNs by securing applications with identity-based policies. It integrates with major identity providers and offers features like multi-factor authentication and logging.
- **OpenZiti** is a game-changer for ZTA networking. It creates secure, overlay networks (zero trust networks) where applications are hidden (dark) and access is granted only after stringent identity and context checks.
- **Cilium** leverages eBPF for highly granular, identity-aware network security policies within Kubernetes environments.
- **Pritunl Zero** is an open source BeyondCorp server that offers zero trust security for privileged access to SSH and web applications. It provides a user-friendly interface and integrates with various identity providers, enabling organisations to implement zero trust principles without